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DIFFERENCE-IN-DIFFERENCES WITH VARIATION IN TREATMENT TIMING

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Difference-in-differences (DD) is both the most common and the oldest quasi-experimental research design, dating back to Snow’s (1855) analysis of a London cholera outbreak.¹ A DD estimate is the difference between the change in outcomes before and after a treatment (difference one) in a treatment versus control group (difference two): $(\bar{y}_{TREAT}^{POST} - \bar{y}_{TREAT}^{PRE}) - (\bar{y}_{CONTROL}^{POST} - \bar{y}_{CONTROL}^{PRE})$. That simple quantity also equals the estimated coefficient on the interaction of a treatment group dummy and a post-treatment period dummy in the following regression:

$$y_{it} = \gamma + \gamma_i TREAT_i + \gamma_t POST_t + \beta^{2x2} TREAT_i \times POST_t + u_{it} \quad (1)$$

The elegance of DD makes it clear which comparisons generate the estimate, what leads to bias, and how to test the design. The expression in terms of sample means connects the regression to potential outcomes and shows that, under a common trends assumption, a two-group/two-period (2x2) DD identifies the average treatment effect on the treated. All econometrics textbooks and survey articles describe this structure,² and recent methodological extensions build on it.³

Most DD applications diverge from this 2x2 set up though because treatments usually occur at different times.⁴ The processes that generate treatment variables naturally lead to variation in timing. Local governments change policy. Jurisdictions hand down legal rulings. Natural disasters strike across seasons. Firms lay off workers. In this case researchers estimate a regression with dummies for cross-sectional units (α_i) and time periods (α_t), and a treatment dummy (D_{it}):

$$y_{it} = \alpha_i + \alpha_t + \beta^{DD} D_{it} + e_{it} \quad (2)$$

¹ A search from 2012 forward of nber.org, for example, yields 430 results for “difference-in-differences”, 360 for “randomization” AND “experiment” AND “trial”, and 277 for “regression discontinuity” OR “regression kink”.

² This includes, but is not limited to, Angrist and Krueger (1999), Angrist and Pischke (2009), Heckman, Lalonde, and Smith (1999), Meyer (1995), Cameron and Trivedi (2005), Wooldridge (2010).

³ Inverse propensity score reweighting: Abadie (2005), synthetic control: Abadie, Diamond, and Hainmueller (2010), changes-in-changes: Athey and Imbens (2006), quantile treatment effects: Callaway, Li, and Oka (forthcoming).

⁴ Half of the 93 DD papers published in 2014/2015 in 5 general interest or field journals had variation in timing.

post-treatment effect, which is closer to -5. My proposed balance test detects significantly higher per-capita income and male/female sex ratios in reform states, in contrast to joint tests of covariate balance across timing groups, which cannot reject the null of balance. Finally, my results show that much of the sensitivity across specifications comes from changes in weights, or a small number of 2x2 DD's, and need not indicate bias.

I. THE DIFFERENCE-IN-DIFFERENCES DECOMPOSITION THEOREM

When units experience treatment at different times, one cannot estimate equation (1) because the post-period dummy is not defined for control observations. Nearly all work that exploits variation in treatment timing uses the two-way fixed effects model in equation (2) (Cameron and Trivedi 2005 pg. 738). Researchers clearly recognize that differences in *when* units received treatment contribute to identification, but have not been able to describe how these comparisons are made.⁹

The simplest way to illustrate how treatment timing works is to consider a balanced panel dataset with T periods (t) and N cross-sectional units (i) that belong to either an untreated group, U ; an early treatment group, k , which receives a binary treatment at t_k^* ; and a late treatment group, ℓ , which receives the binary treatment at $t_\ell^* > t_k^*$. Figure 1 plots this structure.

Throughout the paper I use “group” or “timing group” to refer to collections of units either treated at the same time or not treated. I refer to units that do not receive treatment as “untreated” rather than “control” units because, while they obviously act as controls, treated units do, too. k will denote an earlier treated group and ℓ will denote a later treated group. Each group's sample share is n_k and the share of time it spends treated is $\bar{D}_k$. I use $\bar{y}_b^{POST(a)}$ to denote the sample mean of y_{it} for units in group b during group a 's post period, $[t_a^*, T]$. ($\bar{y}_b^{PRE(a)}$ is defined similarly.)

⁹ Angrist and Pischke (2015), for example, lay out the canonical DD model in terms of means, but discuss regression DD with timing in general terms only, noting that there is “more than one...experiment” in this setting.